

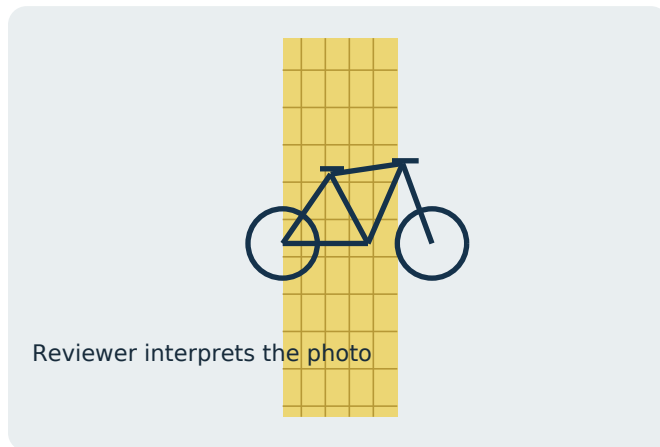
GuidePath — Obstructed tactile paving

ASTLE JOE A S | 13 September 2026 | Detailed companion to the two-page RAP memo

What it does: GuidePath reviews a photograph for tactile paving and supported obstacles, then explains whether a bicycle or motorcycle may overlap the visible guidance surface. Its intended user is a maintenance reviewer who can inspect the site and act.

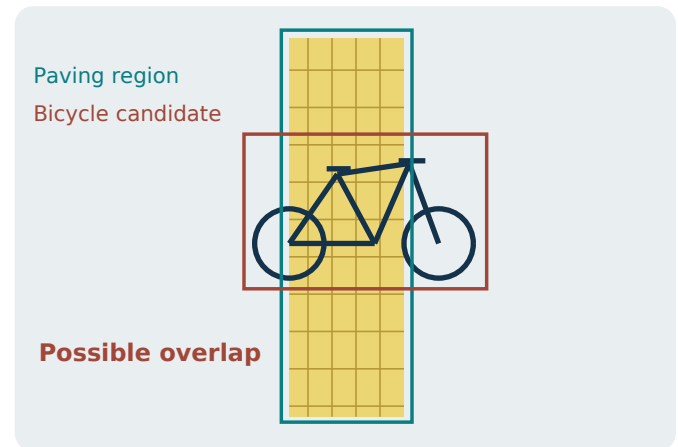
Tactile paving is the textured surface used to help blind and low-vision pedestrians identify guidance or warning features. Existing research already studies its recognition [1, 2]. This project applies that perception task to a limited, evidence-based inspection workflow.

Before / Photo for manual review



Visible tiles and a bicycle need inspection.

After / Proposed evidence view



Same scene; the obstacle has not been removed.

Illustration only. The diagrams above show the same schematic scene before and after proposed AI annotation. They are not photographs, actual model predictions or proof that a route was cleared.

The social purpose

A useful outcome would be helping campus, station or street-maintenance teams notice possible interruptions to tactile paving and review them consistently. The benefit to pedestrians depends on accurate detection, a human response and the obstacle being addressed; the software alone cannot deliver that outcome.

What is distinctive about this approach

The unit of output is an **inspectable finding**: which object, which paving region, what spatial evidence and what uncertainty. A detection score is kept separate from the claim that a path may be obstructed. Unsupported hazards and uncertain images remain explicit limitations.

Status: prototype code, dataset audit and a successful training smoke command are evidenced. Final detection metrics, delivered weights, live LLM verification and deployment remain unverified. Intended impact is not a measured result.

Existing work and my design choices

The contribution is a specific integration and evaluation approach, not a claim of a world-first invention.

Reference point	What it provides	GuidePath adds or proposes
Manual photo review	A human interprets the scene and decides whether a site needs inspection. This is a reference workflow, not a measured study.	A consistent evidence format to support that review; time savings must be tested.
RT-DETR toolkit [3]	Trainable object detection with class labels, boxes and scores.	Custom paving classes plus handwritten relationship and uncertainty rules.
WOTR research dataset [1]	Road scenes with tactile paving and obstacle annotations.	A documented three-class mapping and an audit before accepting the supplied split.
GuideTWSI research [2]	Highlights tactile-surface recognition, viewpoint coverage and geographic variation.	An inspection use case with explicit limits; no claim of better perception or broader coverage.

Decisions and why they matter

1. Choose inspection as the product scope. A photograph cannot establish safe clearance or the entire route. A staff review task is therefore a more defensible first application than pedestrian navigation.

2. Limit the first model to three classes. Tactile paving, bicycle and motorcycle are present in the chosen data. Limiting coverage makes the boundary explainable; it also means the model cannot clear a path of every possible hazard.

3. Change the dataset when the labels do not fit. The earlier candidate had paving classes but lacked the obstacle coverage needed for this approach. WOTR better matched the detection task.

4. Treat data checks as part of modelling. The audit found exact duplicates and invalid boxes. The response was a documented cleanup procedure, not changing the checker to accept questionable data.

5. Make a testable claim. The hypothesis is that spatial overlap reduces false obstruction alarms compared with simply seeing paving and a bike in the same image. That improvement still needs to be measured.

Originality boundary: the contribution is the chosen problem framing, evidence contract, audit decisions and planned comparison. Tactile paving detection, object detectors and confidence thresholds are existing ideas.

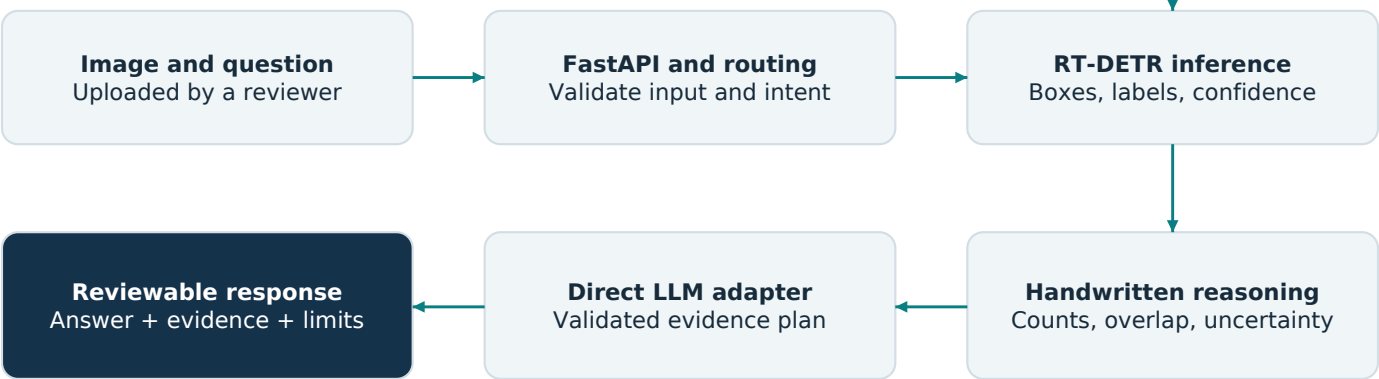
System architecture and technology

Offline training produces a checkpoint; the API uses that checkpoint to answer an inspection question.

OFFLINE / Build and assess the model



API REQUEST / Inspect the visible scene



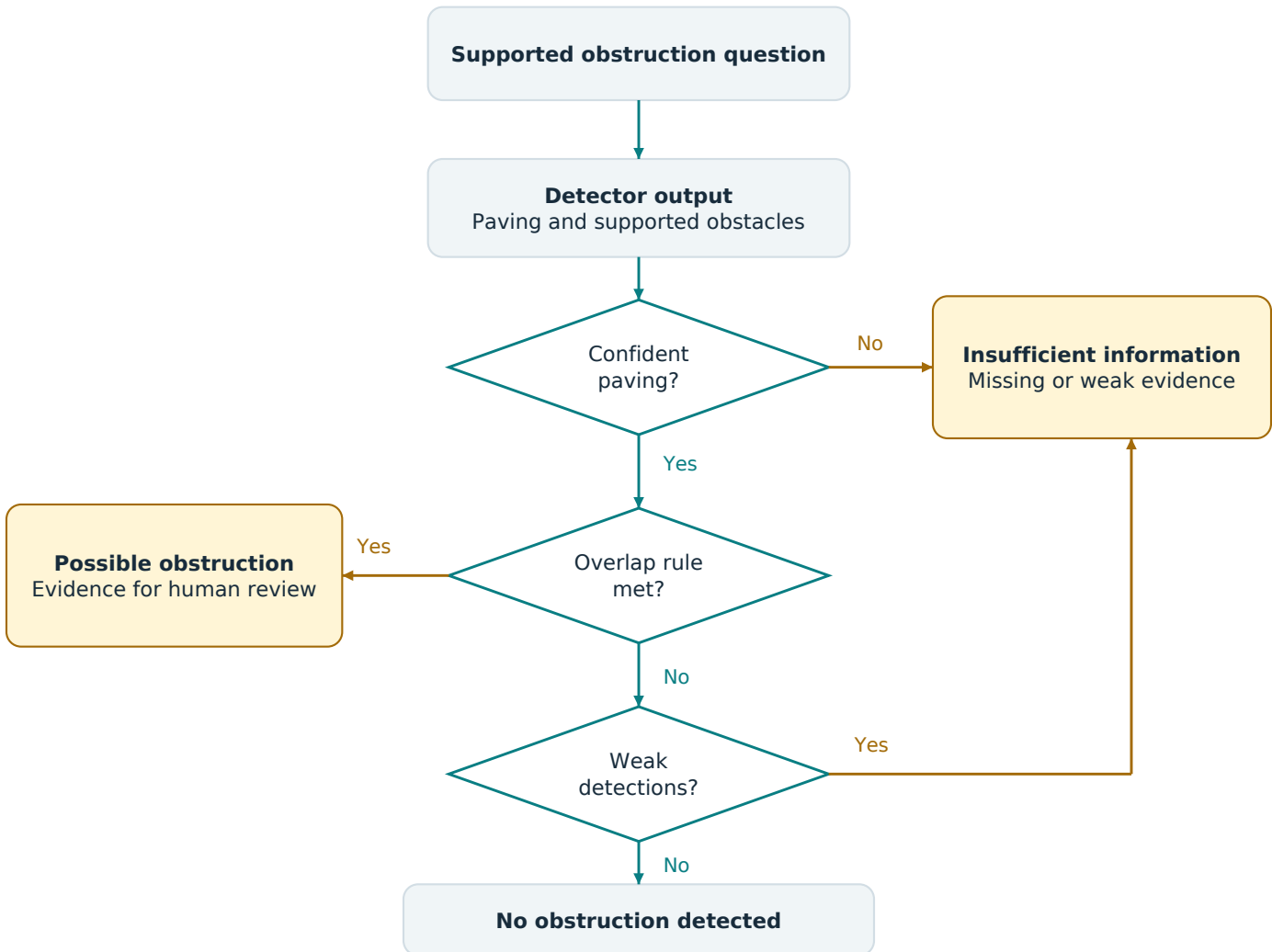
Capability and unsupported questions can skip the detector. The architecture shows the image-dependent path. The LLM selects a conclusion, supporting fact IDs and caveats. Code validates the plan and renders verified facts. A live provider call is still unverified.

Technology	Role in simple terms	Why it is used
Python and PyTorch	Implement the pipeline and run model training.	Transparent code and a supported ML runtime.
Ultralytics RT-DETR-L	Learn where the three selected classes appear.	A faithful RT-DETR implementation allowed by RAP [3].
WOTR and VOC conversion	Turn source labels into the training format.	Makes class mapping and annotation checks explicit.
FastAPI and Pillow	Accept an image and expose two endpoints.	Structured requests plus image decoding and validation.
Handwritten rules and direct LLM call	Calculate evidence and explain its limits.	No agentic framework; facts remain controlled by code.
Colab T4, tests and Docker	Run training, check software and package the API.	T4 and tests were demonstrated; Docker deployment is pending.

Decision flow and a worked example

This flow applies specifically to a supported question about tactile-paving obstruction.

The router first determines the question type. Counts and object lists use their own rules; capability questions skip detection. A request for physical clearance, route safety or unsupported information receives a limitation. An image is required for a visual question.



For image-based answers, the LLM selects relevant evidence IDs and caveats under a verified conclusion; code validates the plan before rendering. A failed or unconfigured LLM call is reported and keeps the deterministic answer. The three result states describe visible evidence only.

Example with illustrative numbers

Suppose a bicycle box has area **4,000 pixels²** and intersects a paving box over **1,000 pixels²**. The overlap fraction is **$1,000 \div 4,000 = 0.25$** . If the detections pass the chosen confidence threshold, this exceeds the current provisional overlap threshold of 0.10.

Answer: “A bicycle appears to overlap the detected paving and may obstruct it. The photograph cannot confirm physical clearance.” The 0.25 fraction describes rectangular image area; it is not a 25% chance of obstruction. All numbers in this example are illustrative.

Data investigation and evaluation

Actual audit findings are separated from proposed cleanup and unmeasured model performance.

The original converter maps **blind_road** to **tactile_paving**, keeps bicycle and motorcycle labels, and retains images without selected classes as background examples. The following is the **pre-cleanup audit**, not final training statistics.

Split	Images	Paving	Bicycles	Motorcycles
Train	9,056	1,596	4,012	8,059
Validation	2,338	371	987	1,887
Test	2,534	409	992	2,202

Before and after the proposed cleanup

Before / observed	After / procedure supplied
34 audit flags including invalid selected-class boxes and exact duplicate images. Flags are not a count of distinct bad images.	Exclude whole images with invalid selected-class boxes; otherwise some visible target objects would be left unlabelled.
Train 10007690 equals validation 30007264. Train 10007030 equals test 20006850.	Keep one identical image, preferring test, then validation, then train; record every exclusion and preserve original outputs.
Source split alone does not establish independent scenes.	Review similar locations and sequences. Exact-image hashes do not detect every near duplicate.

Final cleaned counts were not supplied for this report. The cleanup policy and its report must be checked before final training. If capture groups are available, keep each location/sequence within one split and disclose how the split changed.

How success would be tested

Detection: report per-class mAP@50, mAP@50-95, precision, recall and confusion. Include target counts and tactile-paving recall so a strong bike class cannot conceal weak paving detection.

Reasoning: manually label separate validation and test image sets without using the overlap rule as the ground truth. Compare the spatial rule with a co-occurrence baseline, which flags whenever both paving and a supported obstacle appear. Report false positives, missed visual obstruction candidates and abstentions.

Reproducibility: freeze the dataset manifest, class mapping, checkpoint hash, environment, seed, hardware, training duration and configuration. Select thresholds on validation data, then evaluate the frozen configuration on test data. Hidden-set performance remains unknown.

In simple terms: precision asks how many detections were correct; recall asks how many labelled objects were found. mAP summarizes detection quality across classes and confidence thresholds. None of these establishes whether the physical route is safe.

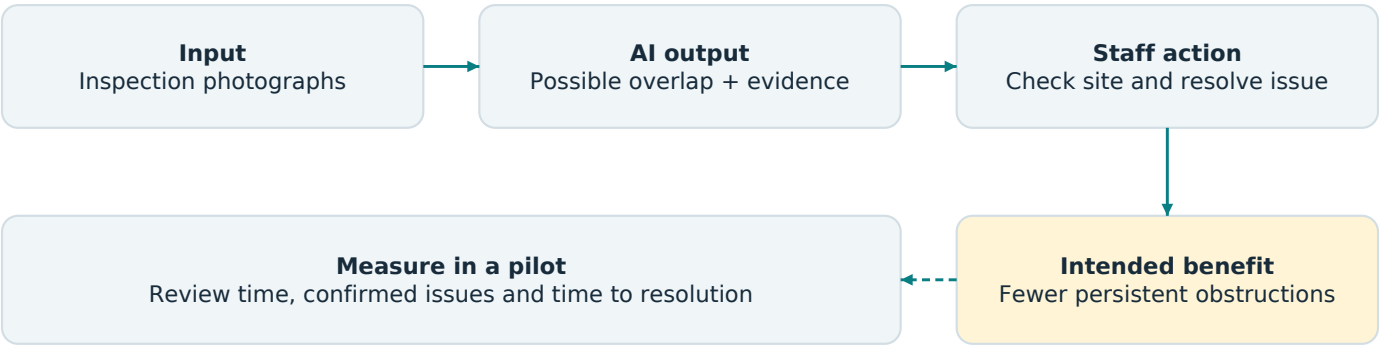
Failure analysis and social impact

Five planned investigations and a measurable path from model output to a useful maintenance action.

Five observed model failures are still required. The rows below are hypotheses to investigate, not measured failures. Each final case must include the image ID, ground truth, actual prediction, evidence for the likely cause and the status of any fix.

Failure hypothesis	Why it could happen	Evidence and response
Object beside diagonal paving	The rectangular paving box includes nearby pavement.	Inspect actual false alarms; consider a paving mask only after validating the need.
Distant paving missed	Few pixels may represent its texture and edges.	Compare misses by target size; test resolution changes on validation data.
Occluded paving	A bike or crowd hides the guidance surface.	Review missing/weak paving; assess whether abstention is appropriate.
Shadow or poor lighting	The visible contrast differs from the training examples.	Inspect originals and labels; test targeted augmentation if supported.
Bike class confusion or duplicate counts	Similar appearance or multiple predictions can alter object counts.	Review matched boxes and counts; measure any proposed suppression change.

How the intended benefit reaches society



Blind and low-vision pedestrians are the intended beneficiaries. Staff must inspect and address each issue. Location-linked maintenance tickets and verified resolution are future features, not implemented capabilities.

Pilot question	Measure before claiming impact
Does it help reviewers?	Median review time and false-alarm workload, compared with manual review on the same photographs.
Does it identify useful cases?	Fraction of flagged locations confirmed by staff; inspect missed cases and abstentions too.
Do physical conditions improve?	Time to resolve confirmed issues and a follow-up site inspection. No reduction is claimed yet.

RAP requirements and delivery evidence

Use the two-page memo for the required memo field; this longer report is supplementary material.

RAP component	%	Evidence and present status
Hidden-set detection	25%	Custom RT-DETR training path supplied. Final checkpoint and hidden performance are unverified.
Dataset sourcing and labels	15%	WOTR source, mapping, original class counts and 34 audit flags are documented. Final cleanup remains to verify.
Evaluation methodology	10%	Evaluation scripts and split policy exist. Final measured detection and reasoning results are pending.
Five failure cases	15%	Exporter and five investigation hypotheses exist. Five actual model cases are still required.
Reasoning and uncertainty	15%	Routing and evidence logic were tested. Typed evidence-plan synthesis is implemented; live LLM verification is pending.
Code and reproducibility	10%	Training/evaluation/API code and environment logging supplied. 29 software tests passed.
Bonus engineering	10%	Dockerfile, logging and error handling supplied. Container build and public deployment are unverified.

Current evidence and next deliverables

Colab showed a T4 and a successful one-epoch smoke command. Next: preserve a usable checkpoint and run record, then produce metrics, failure cases, a live API/LLM check and deployment.

API usage: POST /detect takes an image and returns labels, boxes and confidence. POST /reason takes an image and question; capability questions can omit the image. Interactive local documentation is at /docs. Example: “Is the tactile path blocked?” An image with no confident paving should produce insufficient information.

What this work demonstrates

The candidate must explain the dataset pivot, duplicate findings, invalid-label policy, limited class coverage and geometry limits. AI assisted the implementation; the candidate must understand and defend the choices and code.

Sources and evidence notes

[1] Xia et al. A dataset for the visually impaired walk on the road, Displays 79 (2023), 102486. Author repository: github.com/kxZR/WOTR.

[2] GuideTWSI, arXiv:2603.07060 (2026). Primary research on tactile walking surface data and generalization: arxiv.org/abs/2603.07060.

[3] Ultralytics official RT-DETR documentation: docs.ultralytics.com/models/rtdetr. Public sources checked 13 September 2026.

[4] Project evidence: RAP brief and email, shared audit, Colab screenshots and software test report.